TDF Radial Reconstruction on a Preregistered Controlled SPARC Subset: Holdout Validation, Failure Modes, and Sensitivity-Recovery Cases

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Abstract

We benchmark a Time-Delay Field (TDF) radial reconstruction framework on a pre-registered, controlled subset of SPARC rotation curves [6]. This is *not* full-SPARC validation: results apply to the expansion-20 cohort (n=20) and the nested expansion-12 cohort (n=12). The primary model is `tdf_3knot`; `tdf_5knot` is a sensitivity/high-flexibility diagnostic only. We use train-only even/odd holdout comparisons against NFW [10] and MOND [9] baselines at fixed canonical baryons and fixed K_g (legacy benchmark notation K_τ). In the pre-registered controlled expansion-20 cohort, the primary conservative `tdf_3knot` model achieves robust holdout success in 15 of 20 galaxies. Three additional galaxies show sensitivity-recovery where `tdf_5knot` improves substantially but is not counted as primary success. NGC7814 remains the only all-TDF holdout failure, and UGC00128 remains a mixed near-tie case. The analysis does not disprove dark matter, does not replace ΛCDM, does not test lensing, claims no universal τ-profile, and makes no final M/L calibration claim.

1 Introduction

Galaxy rotation curves provide a disciplined test of how well models reproduce observed circular speeds beyond the baryonic contribution [6]. This paper is a *controlled benchmark* of TDF as a reconstruction framework—not a dark-matter-disproof or ΛCDM-replacement study. Predictive holdout validation matters because in-sample fits can reward extra flexibility; we therefore require `tdf_3knot` to perform on held-out radii against standard baselines before counting primary success. We extend an earlier six-galaxy pilot benchmark (repository history) that established an explicit NGC7814 failure mode to pre-registered expansion-12 and expansion-20 cohorts with frozen failure-mode taxonomy [7].

Many rotation-curve studies optimize in-sample fit quality. Here we prioritize predictive holdout consistency to reduce flexibility-driven overfitting [5, 2]. We do not claim that holdout success "proves" TDF at cosmological level; it is a conservative empirical gate on the controlled subset.

2 TDF reconstruction framework

TDF reconstructs a radial $\tau(r)$ from rotation-curve residuals relative to fixed SPARC baryonic components [6, 8].

2.1 Benchmark definition

We decompose the observed rotation curve into baryonic and TDF-mediated contributions,

$$v_{\rm obs}^2(r) = v_{\rm bar}^2(r) + v_\tau^2(r), \tag{1}$$

with fixed catalog baryons $v_{\rm bar}^2 = v_{\rm gas}^2 + v_{\rm disk}^2 + v_{\rm bulge}^2$. The TDF closure used in this benchmark is

$$v_\tau^2(r) = r\,K_g\,\frac{d\tau}{dr}, \tag{2}$$

where K_g is the fixed gravitational projection coefficient held constant across the cohort. Frozen repository configs and benchmark tables use the legacy label K_τ for the same projection role; numerical values are unchanged. The dynamical stiffness κ_τ in the mother field equation is *not* the same parameter as K_g and is not varied here. Given $v_{\rm obs}(r)$ and $v_{\rm bar}(r)$, the direct reconstruction law is

$$\frac{d\tau}{dr} = \frac{v_{\rm obs}^2(r) - v_{\rm bar}^2(r)}{r\,K_g}. \tag{3}$$

Equation (3) defines a *galaxy-specific* $\tau(r)$ profile for each system. The algebraic reconstruction law is universal; we do **not** claim a universal functional $\tau(r)$ shape across galaxies. Repository diagnostic pointwise reconstruction from Eq. (3) is not used as a primary fitted competitor.

2.2 Fitted knot regularization

Fitted models (`tdf_3knot`, `tdf_5knot`) are low-parameter piecewise-linear regularizations of $d\tau/dr$ with knot amplitudes optimized on training radii during holdout evaluation. `tdf_3knot` is the **primary conservative** parameterization; `tdf_5knot` increases flexibility and is reported only as a sensitivity diagnostic.

3 Controlled SPARC benchmark protocol

The pipeline ingests SPARC `rotmod` data, applies the frozen expansion plan, reconstructs τ, fits NFW/Burkert/MOND baselines, fits TDF knot models, runs holdout validation, audits failure modes, and reconciles claims (Figure 1). All expansion_20 numbers are taken from frozen pipeline outputs; this editing phase does not rerun fits or alter tables.

4 Data and cohort selection

Galaxies enter expansion_20 in a pre-registered order (Table 1). expansion_12 is the nested twelve-galaxy subset used for intermediate auditing; expansion_20 is the main reported cohort. Membership is fixed in `sparc_subset_expansion_plan.csv` with no post-hoc cherry-picking.

5 Baseline models

5.1 Baryonic-only

$v_{\rm bar}^2 = v_{\rm gas}^2 + v_{\rm disk}^2 + v_{\rm bulge}^2$ from catalog inputs without fitting M/L in the primary benchmark.

5.2 NFW and Burkert

NFW [10] and Burkert [3] halos use log-space multistart refits; boundary-limited solutions and high reduced χ^2 may occur and should not be overinterpreted.

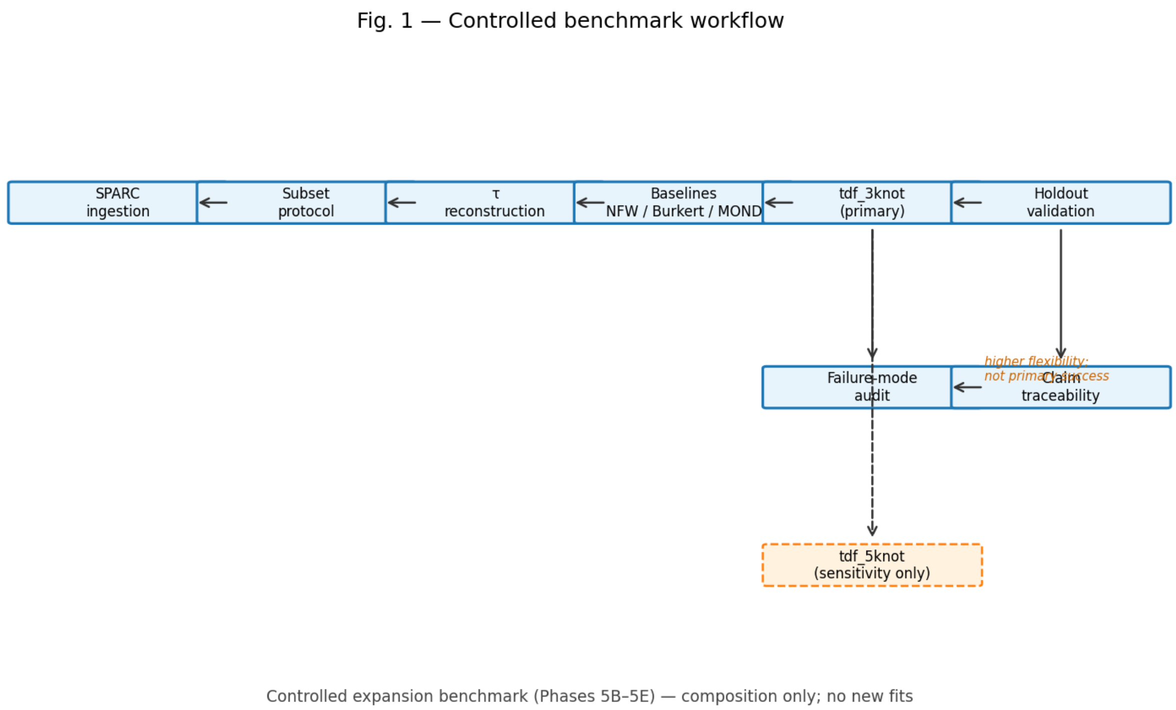

Figure 1: End-to-end benchmark workflow (ingestion through claim reconciliation). The solid path is the primary conservative pipeline (`tdf_3knot`); the dashed branch is the high-flexibility sensitivity pipeline (`tdf_5knot`), which is audited separately and excluded from primary success counts.

Table 1: Pre-registered expansion_20 cohort membership (Phase 5A plan).

Galaxy	Role	In expansion_12	Order
DDO154	original controlled six	Y	1
IC2574	original controlled six	Y	2
NGC2403	original controlled six	Y	3
NGC3198	original controlled six	Y	4
NGC6503	original controlled six	Y	5
NGC7814	original controlled six	Y	6
UGC02953	expansion addition	Y	7
UGC05253	expansion addition	Y	8
UGC09133	expansion addition	—	9
UGC06787	expansion addition	—	10
NGC5055	expansion addition	Y	11
UGC00128	expansion addition	Y	12
NGC0289	expansion addition	Y	13
UGC12506	expansion addition	—	14
UGC11455	expansion addition	—	15
NGC6015	expansion addition	—	16
DDO161	expansion addition	Y	17
NGC7793	expansion addition	—	18
UGC07524	expansion addition	—	19
UGC08490	expansion addition	—	20

Source: `outputs/tables/sparc_subset_expansion_plan.csv`.
Caveat: Controlled subset only; not full-SPARC validation.

5.3 MOND and RAR

MOND [9, 4] with fitted a_0 and optional RAR [8] are empirical rotation-curve baselines only, not cosmological or lensing validations of MOND.

Burkert baseline scope. Burkert [3] halos are implemented for completeness but are not emphasized in the primary holdout gate because, under the frozen log-space multistart halo protocol, Burkert fits frequently exhibit boundary-limited parameters and high reduced χ^2 on this subset. NFW [10] and MOND [9, 4] are the headline non-TDF comparators for `robust_tdf_success`; Burkert results remain in pipeline tables for reproducibility.

6 Holdout validation methodology

Primary split: even/odd index holdout with train-only refit. **Classification definitions (expansion_20):**

- **robust_tdf_success:** `tdf_3knot` beats both NFW refit and MOND fit-a_0 on even/odd holdout RMSE (primary success).
- **sensitivity_recovery:** `tdf_3knot` fails the primary gate but `tdf_5knot` improves substantially (not primary success).
- **tdf_failure_mode:** all TDF knot variants fail the primary holdout gate (NGC7814).
- **mixed_result:** near-tie among models without a clear primary win (UGC00128).

Additional splits (blocked radial thirds, k-fold) are diagnostic [12, 2]. Model parsimony is summarized with AIC/BIC in pipeline tables [1, 11]; the primary gate here is predictive holdout RMSE, not in-sample information criteria alone. Table 2 lists per-galaxy even/odd RMSE.

Table 2: Even/odd holdout test RMSE [km/s] for expansion_20 (train-only refit).

Galaxy	Classification	`tdf_3knot`	NFW	MOND	`tdf_5knot`
DDO154	Robust TDF success	2.03	3.56	3.98	1.44
DDO161	Robust TDF success	0.72	3.87	6.55	0.78
IC2574	Robust TDF success	2.98	6.39	6.29	1.73
NGC0289	Robust TDF success	19.81	21.91	22.95	19.54
NGC2403	Robust TDF success	8.66	9.76	14.24	3.44
NGC3198	Robust TDF success	10.45	12.00	14.34	8.24
NGC5055	Sensitivity recovery	142.86	56.87	55.37	19.56
NGC6015	Robust TDF success	7.13	8.27	13.45	4.86
NGC6503	Robust TDF success	9.33	10.14	12.39	4.90
NGC7793	Robust TDF success	4.57	8.15	8.47	3.80
NGC7814	TDF failure mode	155.80	24.89	27.85	113.25
UGC00128	Mixed result	3.00	2.72	6.24	5.06
UGC02953	Robust TDF success	37.80	41.40	44.21	26.25
UGC05253	Sensitivity recovery	48.52	36.15	37.38	16.46
UGC06787	Robust TDF success	38.29	38.58	43.09	30.78
UGC07524	Robust TDF success	1.63	2.82	3.58	1.52
UGC08490	Robust TDF success	1.13	1.57	1.82	1.15
UGC09133	Robust TDF success	33.24	36.27	37.96	25.74
UGC11455	Robust TDF success	15.81	39.35	39.40	15.97
UGC12506	Sensitivity recovery	11.92	8.00	20.14	5.76

Source: `expansion20_failure_mode_summary.csv`.
Caveat: Primary gate uses `tdf_3knot`; `tdf_5knot` is sensitivity only.

6.1 Descriptive holdout statistics

Descriptive summary (expansion_20, even/odd holdout; not hypothesis testing): robust primary success: 15/20 (75%; descriptive 95% bootstrap interval [55%, 90%]); sensitivity-recovery: 3/20 (15%; descriptive 95% bootstrap interval [0%, 30%]); TDF failure: 1/20 (5%; descriptive 95% bootstrap interval [0%, 15%]); mixed: 1/20 (5%; descriptive 95% bootstrap interval [0%, 15%]). Primary beats NFW: 15/20 (75%; descriptive 95% bootstrap interval [55%, 90%]); primary beats MOND: 17/20 (85%; descriptive 95% bootstrap interval [70%, 100%]). Median $\Delta\mathrm{RMSE} \equiv \mathrm{RMSE}_{3\mathrm{k}} - \mathrm{RMSE}_{\mathrm{baseline}}$: -1.17 km/s vs NFW (galaxy-level 2.5–97.5% range [-14.07, 109.57] km/s) and -3.60 km/s vs MOND ([-16.29, 108.74] km/s). Negative ΔRMSE indicates lower primary TDF holdout error. We report bootstrap percentile intervals for cohort fractions as *descriptive* uncertainty brackets only; they are not hypothesis tests and do not establish cosmological significance.

7 Results

7.1 expansion_20 (main cohort)

In the pre-registered controlled expansion_20 cohort, the primary conservative `tdf_3knot` model achieves robust holdout success in 15 of 20 galaxies. Three additional galaxies show sensitivity-recovery where `tdf_5knot` improves substantially but is not counted as primary success. NGC7814 remains the only all-TDF holdout failure, and UGC00128 remains a mixed near-tie case. Table 3 summarizes cohort-level counts. Figure 2 contrasts expansion_12 and expansion_20 classifications. Representative robust-success rotation curves (subset examples only) appear in Figure 3.

Table 3: Classification and holdout comparison: expansion_12 vs expansion_20.

Metric	expansion_12	expansion_20	Notes
Cohort size	12	20	expansion_12 expansion_20 (same original six + overlapping additions)
Robust TDF success (primary)	8	15	Primary conservative success (tdf_3knot beats NFW and MOND on holdout for e20)
Sensitivity recovery	0	3	Phase 5C class; not counted as primary success in e20
TDF failure mode	2	1	NGC7814 all-TDF failure in both cohorts
Mixed result	2	1	UGC00128 near-tie in both
Primary beats NFW (holdout)	8	15	even/odd holdout; train-only refit
Primary beats MOND (holdout)	9	17	Required for robust_tdf_success in expansion_20

Source: `controlled_expansion_comparison_summary.csv`.
Caveat: Robust TDF success requires primary `tdf_3knot` to beat both NFW and MOND on holdout.

7.2 Primary success counts and Table 2 interpretation

Primary success is defined only through `tdf_3knot`: 15/20 robust successes. Seventeen galaxies show `tdf_3knot` beating MOND and fifteen beating NFW on even/odd holdout, but robust success requires **both**. Table 2 shows a heterogeneous distribution: most robust-success galaxies exhibit primary RMSE at or below NFW/MOND, while **NGC7814** is a decisive counterexample where all TDF variants remain poor on holdout despite competitive in-sample flexibility. Three `sensitivity_recovery` galaxies (NGC5055, UGC05253, UGC12506) illustrate why `tdf_5knot`

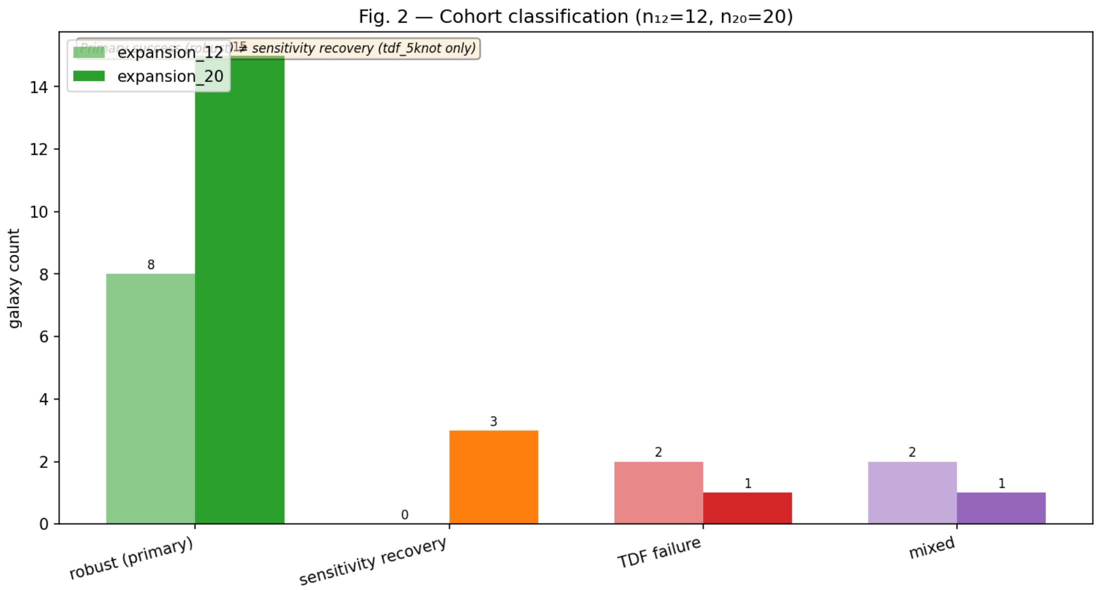

Figure 2: Failure-mode counts for expansion-12 (n=12) and expansion-20 (n=20). **Robust TDF success** (primary, `tdf_3knot`) is shown separately from **sensitivity-recovery** (`tdf_5knot` diagnostic only).

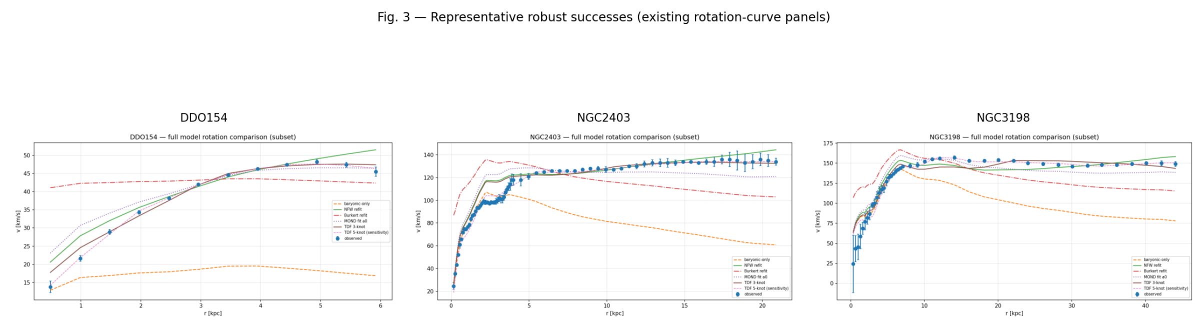

Figure 3: Representative robust-success examples (DDO154, NGC2403, NGC3198). Panels are illustrative subset cases only, not a complete census of successes.

is reported separately: large holdout gains can reflect added knot freedom rather than validated primary-model superiority. We do not extrapolate cohort fractions to the full SPARC catalog.

7.3 expansion_12 (nested context)

expansion-12 (n=12) is a nested audit cohort: eight robust primary successes before the expansion-20 taxonomy refinements (e.g. NGC5055 reclassified as sensitivity-recovery in expansion-20). We report expansion-12 for continuity but treat expansion-20 as the main result.

8 Failure modes and sensitivity recovery

NGC7814 remains the sole canonical all-TDF holdout failure (Figure 4). **NGC5055**, **UGC05253**, and **UGC12506** exhibit sensitivity-recovery: `tdf_5knot` reduces holdout error markedly but these are **not** primary successes (Figure 5). **UGC00128** is a mixed near-tie (NFW marginally best). Per-galaxy holdout RMSE for all twenty galaxies is shown in Figure 6. Table 4 lists the five non-robust cases.

Table 4: Non-robust expansion_20 cases: holdout diagnostics (five galaxies).

Galaxy	Classification	$RMSE_{3k}$	$RMSE_{5k}$	ΔRMSE [km/s]
NGC7814	TDF failure mode	155.80	113.25	42.55
NGC5055	Sensitivity recovery	142.86	19.56	123.31
UGC05253	Sensitivity recovery	48.52	16.46	32.05
UGC12506	Sensitivity recovery	11.92	5.76	6.16
UGC00128	Mixed result	3.00	5.06	-2.06

Source: `expansion20_failure_diagnostics.csv`.
Caveat: Descriptive labels only; not causal claims about baryons or dark matter.

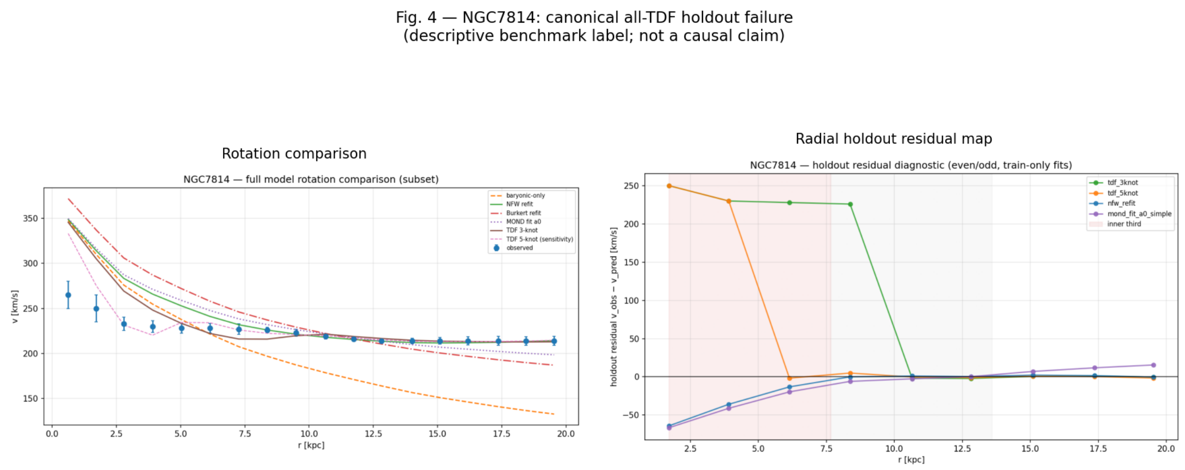

Figure 4: NGC7814 canonical all-TDF even/odd holdout failure. This is a *descriptive* benchmark label for external reporting, not causal proof about baryons, halos, or dark matter.

9 Limitations

- **Scope:** controlled expansion_20 only—not full-SPARC validation.
- **Baryons and M/L:** fixed canonical decomposition; no final M/L calibration claim.

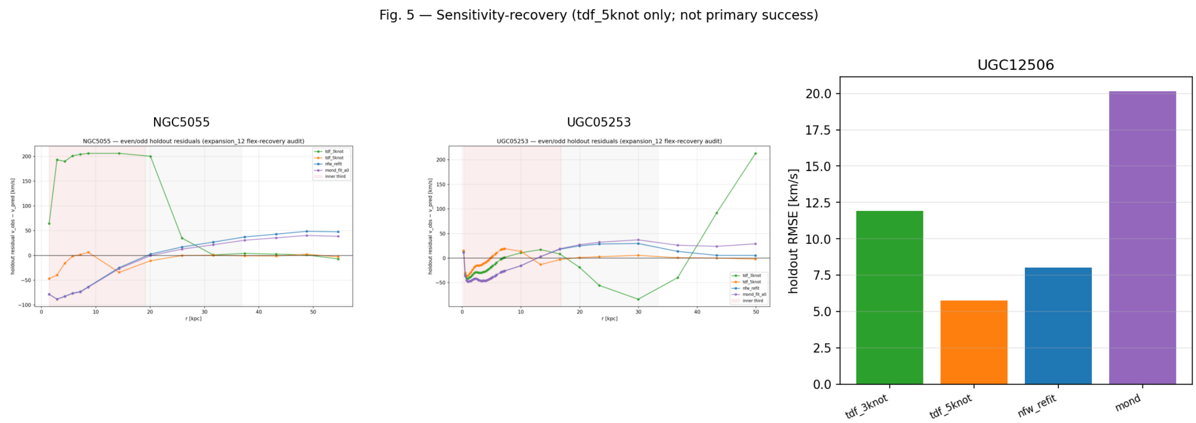

Figure 5: Sensitivity-recovery galaxies (NGC5055, UGC05253, UGC12506). Improved `tdf_5knot` holdout performance is shown for diagnosis only and is **not** counted toward the 15/20 primary robust-success total.

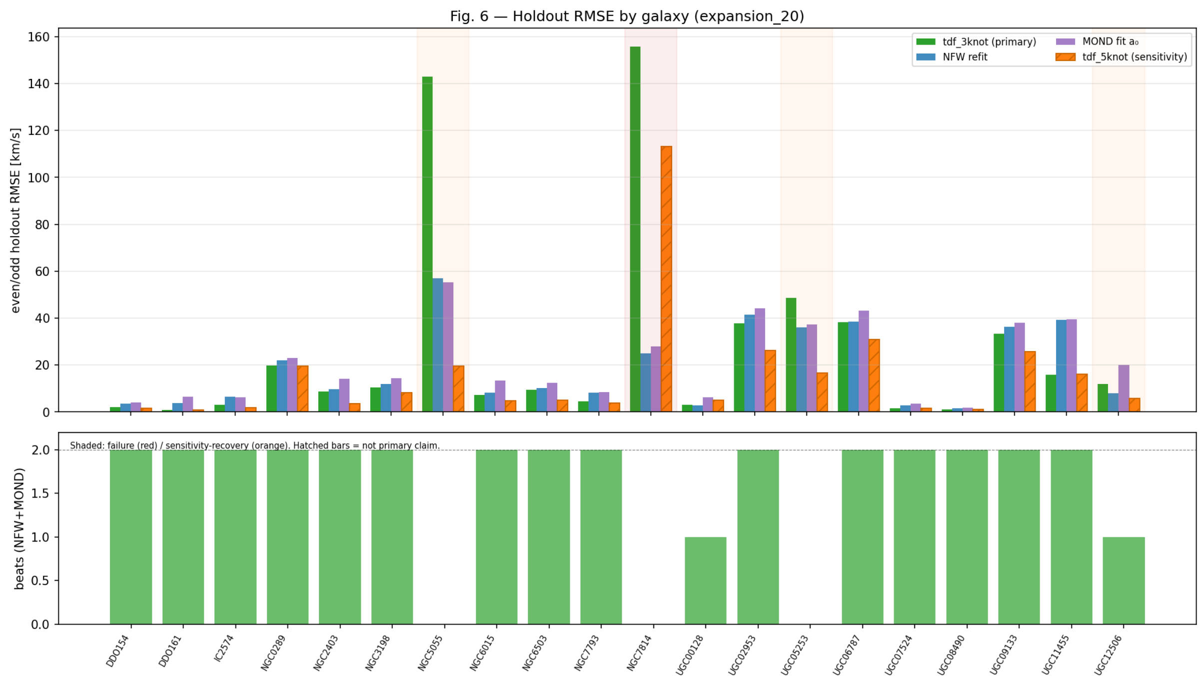

Figure 6: Expansion-20 even/odd holdout RMSE by galaxy. Solid bars: `tdf_3knot` (primary), NFW, and MOND; **hatched** `tdf_5knot` bars are sensitivity-only and not used for primary claims.

- **K_g (legacy K_τ):** gravitational projection coefficient held fixed; not measured or inferred; distinct from field stiffness κ_τ.
- **Cosmology and lensing:** does not disprove dark matter; does not replace ΛCDM; lensing is not tested.
- **Flexibility:** `tdf_5knot` carries higher overfitting/flexibility risk and is sensitivity-only.
- **τ universality:** per-galaxy diagnostics only; no universal τ-profile claim.

Table 5: Assumptions and limitations for the controlled expansion-20 manuscript.

Assumptions (expansion-20)	**Limitations (expansion-20)**
• Benchmark scope is the pre-registered controlled expansion-20 cohort (n=20) only.	• Not full-SPARC validation; cohort fractions must not be extrapolated.
• Fixed canonical SPARC baryonic decomposition; no final M/L calibration in this manuscript.	• Halo and MOND baselines remain degenerate with baryonic assumptions.
• K_g is fixed across the cohort (legacy benchmark label K_τ in frozen tables); not measured or inferred; distinct from field stiffness κ_τ.	• Higher knot count (`tdf_5knot`) increases flexibility and overfitting risk.
• NFW and MOND are rotation-curve empirical baselines, not cosmology or lensing tests.	• Does not disprove dark matter and does not replace ΛCDM.
• Burkert halos are fitted for completeness but are not part of the primary holdout gate.	• Lensing is not tested in this benchmark phase.
• Primary model: `tdf_3knot`; sensitivity model: `tdf_5knot` (not primary success).	• No universal τ-profile claim; per-galaxy τ diagnostics only.
• Even/odd train-only holdout is the primary predictive gate.	• Bootstrap intervals in the manuscript are descriptive, not significance tests.

Source: Phase 5F-F expansion-20 checklist (replaces legacy six-galaxy assumption bullets).
Caveat: Aligned with claims C20-A–C20-H; no change to frozen benchmark metrics.

10 Future work

Full-catalog processing, lensing tests, two-dimensional τ maps, and photometry-calibrated M/L priors require protocol amendments. Blocked-holdout maps for UGC12506 may localize radial failure without changing frozen primary labels.

11 Conclusion

On the pre-registered controlled expansion_20 subset, `tdf_3knot` achieves robust even/odd holdout success in 15 of 20 galaxies at fixed baryons and K_g, with three sensitivity-recovery cases, one all-TDF failure (NGC7814), and one mixed near-tie (UGC00128). TDF is a benchmarked reconstruction framework within explicit claim boundaries, not a cosmology-replacement or dark-matter-disproof result.

A Reproducibility appendix

Benchmark: `python3 scripts/run_expansion20_pipeline.py`; audit: `python3 scripts/build_controlled` tables: `python3 scripts/export_paper_tables.py`; figures: `python3 scripts/build_paper_figures.py`; PDF: `python3 scripts/compile_paper_pdf.py`. Repository: [7].

B Claim-boundary appendix

Table 6 and Figure 7 document claims C20-A–C20-H. Prohibited external language includes full-SPARC validation, dark-matter disproof, ΛCDM replacement, lensing confirmation, and universal τ-profile discovery.

Table 6: Claim traceability matrix for controlled expansion_20 (C20-A–C20-H).

ID	Claim	Status
C20-A	expansion_20 cohort processed reproducibly through frozen pipeline.	Supported
C20-B	Primary tdf_3knot robust holdout success in 15 of 20 galaxies.	Supported (caveat)
C20-C	tdf_5knot improves sensitivity-recovery cases.	Sensitivity only
C20-D	NGC7814 remains all-TDF holdout failure.	Supported
C20-E	TDF validated on full SPARC.	Not supported
C20-F	Dark matter is disproven.	Prohibited
C20-G	Lensing confirms TDF.	Not tested
C20-H	Universal -profile discovered.	Not supported

Source: `controlled_expansion_final_claims.csv`.
Caveat: Authoritative language boundary for external reporting.

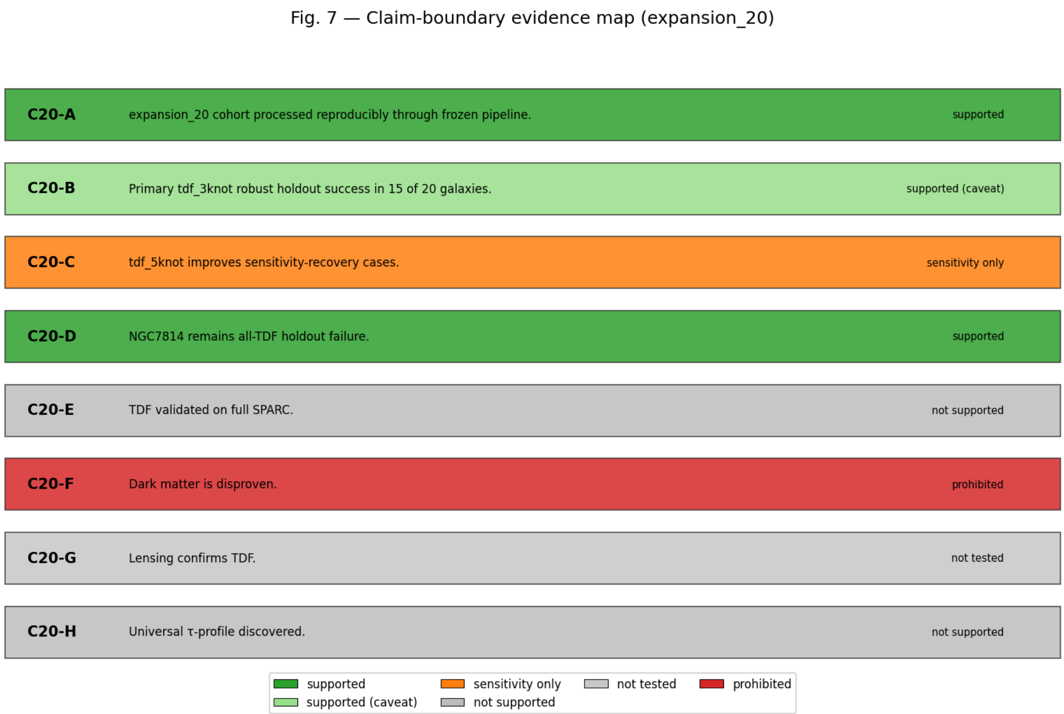

Figure 7: External claim-boundary map for audited claims C20-A–C20-H (supported, caveat, sensitivity-only, not supported, prohibited).

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